

# Data Visualization Guide

## 1. Common Types of Data Visualizations

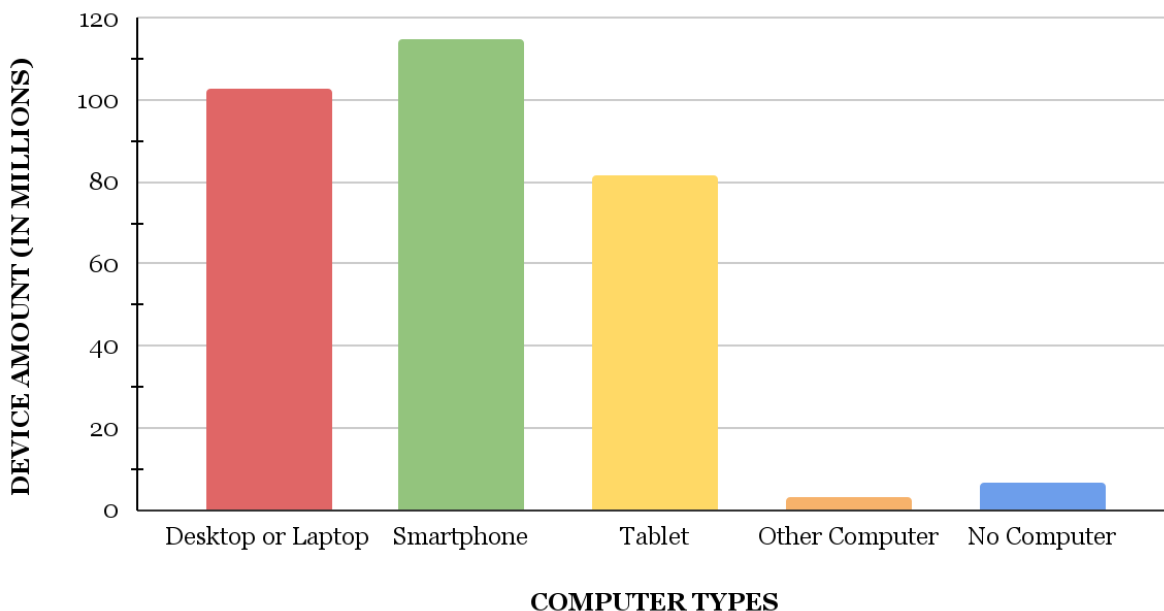
### 1.1 Bar Chart

One of the most common and highly recognizable types of charts, a bar chart shows data values using rectangular bars. The length and breadth of the bars correspond to the magnitude of the data. Bar charts can be vertical or horizontal, depending on the data and space available.

- **When to Use:** Bar and column charts are best for comparing categories or showing changes over time when data is discrete. They are also useful in ranking items and frequency distribution. You can find stacked bar charts in financial reporting, demographic comparisons, surveys, etc.
- **Pros:** Easy to understand; great for comparisons.
- **Cons:** Can become cluttered with too many categories.

### TYPES OF COMPUTERS IN U.S. HOUSEHOLDS (2021)

Data From: United States Census Bureau



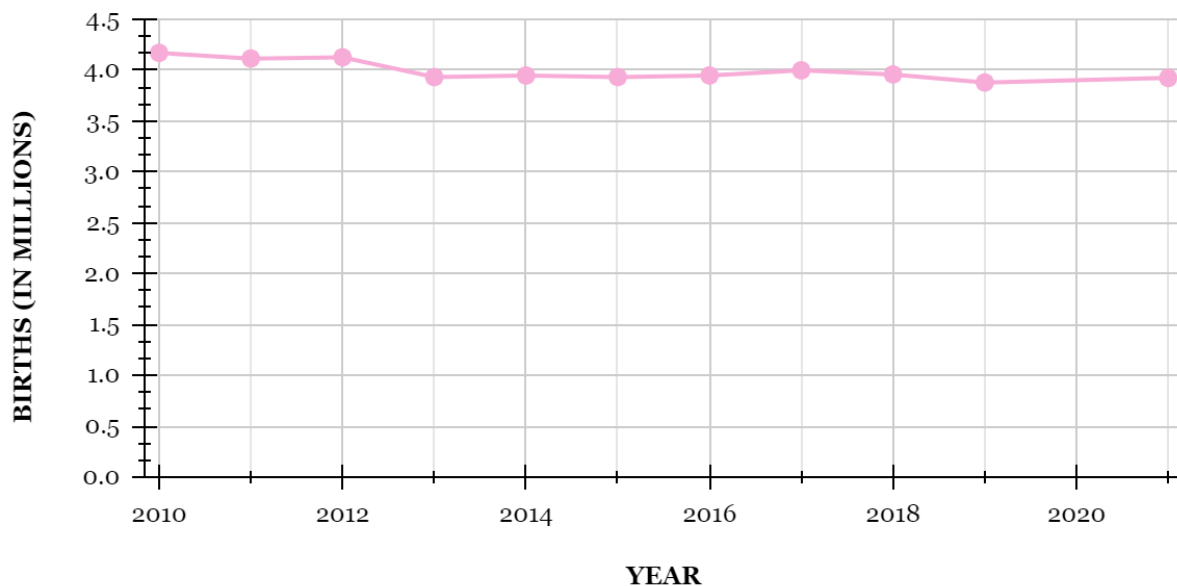
## 1.2 Line Chart

A line chart **shows trends over time** by connecting individual data points in a line. Line graphs are best used with continuous data where the relationships between data points are essential to understanding patterns. A line chart makes it easy for viewers to see upward or downward trends.

- **When to Use:** Line charts are perfect for tracking trends, such as stock market movements, sales growth, or temperature changes over time. They are useful for comparing multiple data sets within the same chart, such as multiple product sales over a year.
- **Pros:** Ideal for continuous data; good for spotting trends.
- **Cons:** Not suitable for categorical data.

### TOTAL BIRTHS IN THE UNITED STATES (2010-2021)

Data From: United States Census Bureau



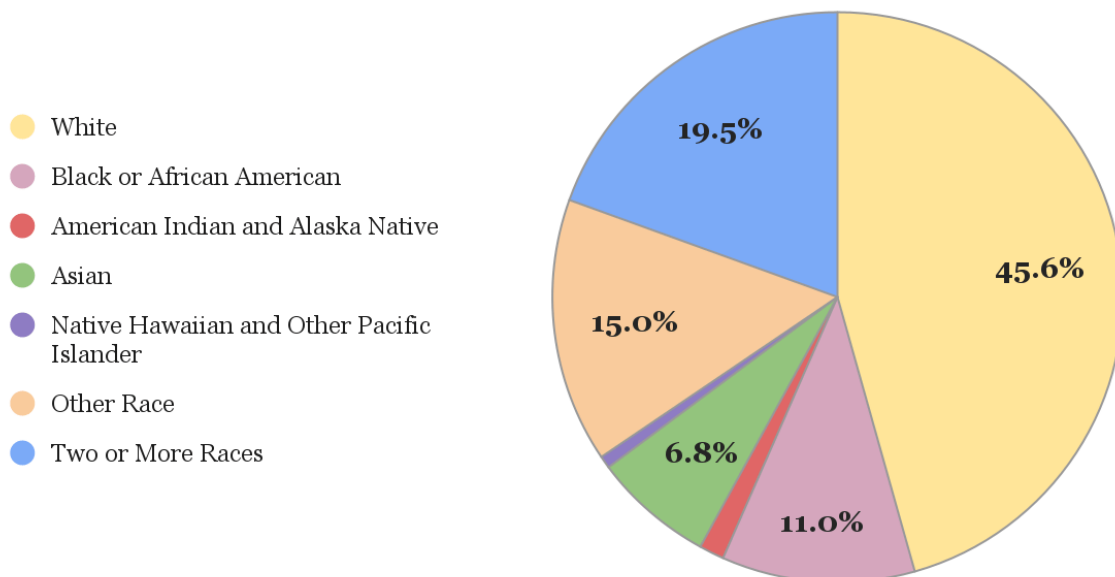
## 1.3 Pie Chart

A pie chart is another classic chart. It is circular in shape and divided into parts. Each part shows the proportion of the category it represents to the total. Pie charts are handy for **showing the percentage each category contributes to the total**.

- **When to Use:** Pie charts are best when you want to show how a whole is divided into parts, like how a company's budget is split between departments or how people responded to a survey in percentages.
- **Pros:** Simple visual for proportions.
- **Cons:** Hard to compare similar sizes; not great with many segments.

### RACE DISTRIBUTION IN LAS VEGAS, NV (2020)

Data From: United States Census Bureau

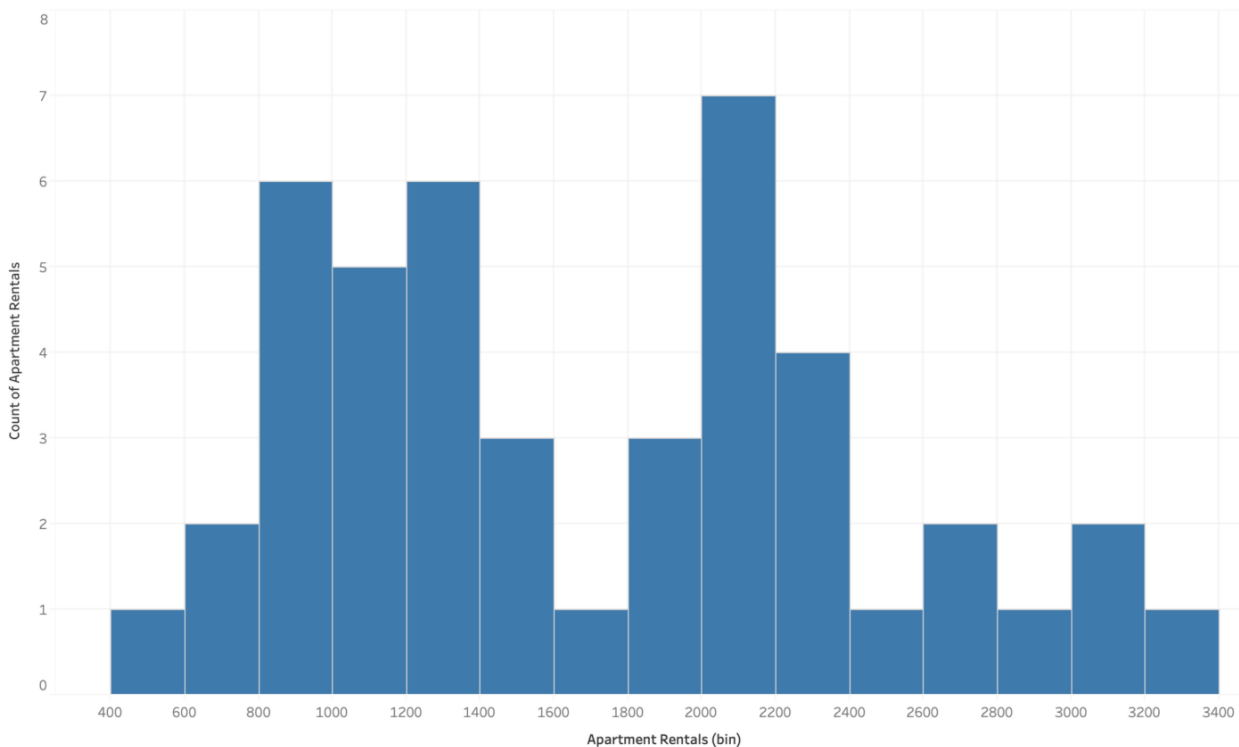


## 1.4 Histogram

Similar to a bar chart, a histogram **shows data distribution over a range of values**. Each bar represents a group of data points, showing how often data falls within that range. But unlike bar charts, the bars in a histogram touch each other to represent continuous data.

- **When to Use:** Histograms are useful for understanding the distribution of data and data preprocessing, such as showing the frequency of test scores, ages in a population, or product defects within certain ranges.
- **Pros:** Useful for identifying skewness or outliers.
- **Cons:** Bin size can mislead; not for comparisons.

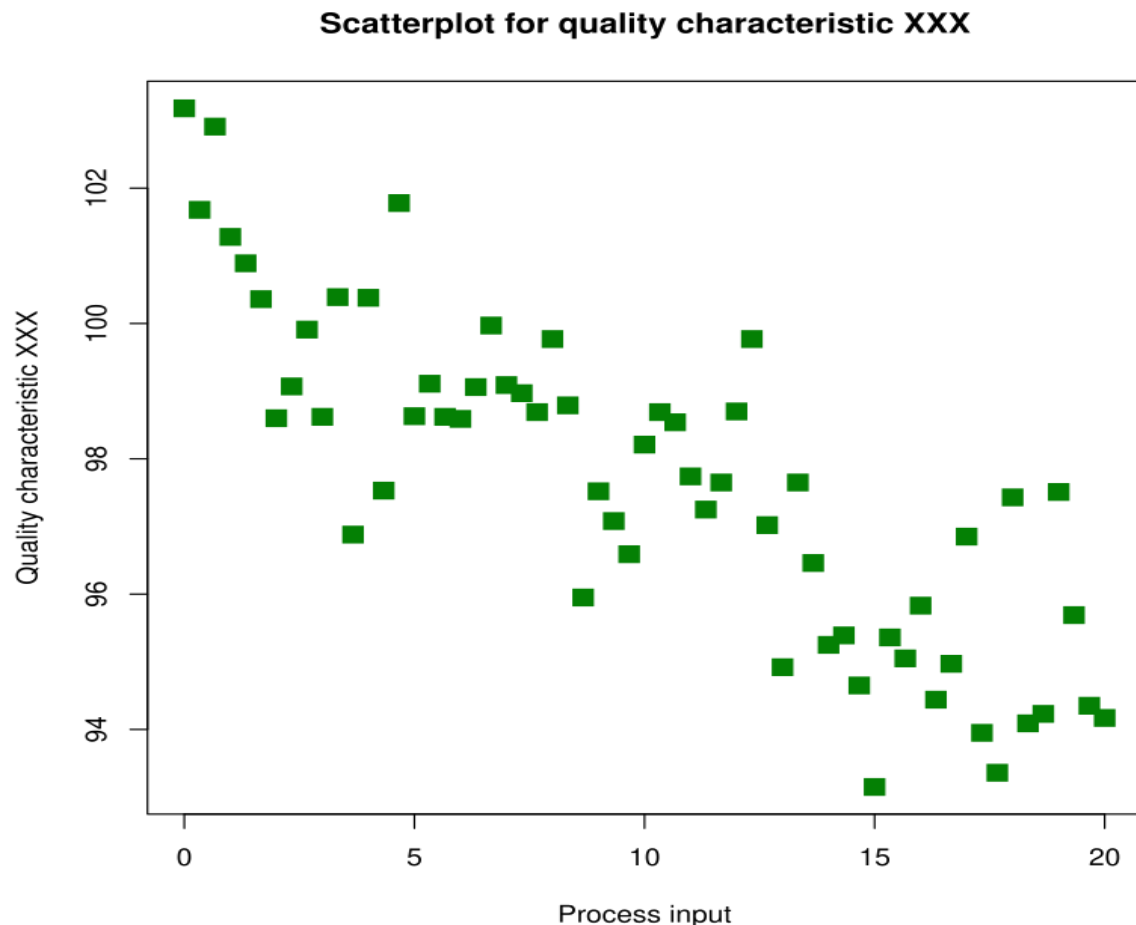
Apartment Rental Prices



## 1.5 Scatter Plot

A scatter plot uses dots to **represent the relationships between two numeric variables**. Each dot represents an observation, and its position is determined by the two variables' values on the vertical axis and horizontal axis. So, if datapoints are close on the scatter chart, they are more related. On the other hand, the farther they are, the less connected their relationship is.

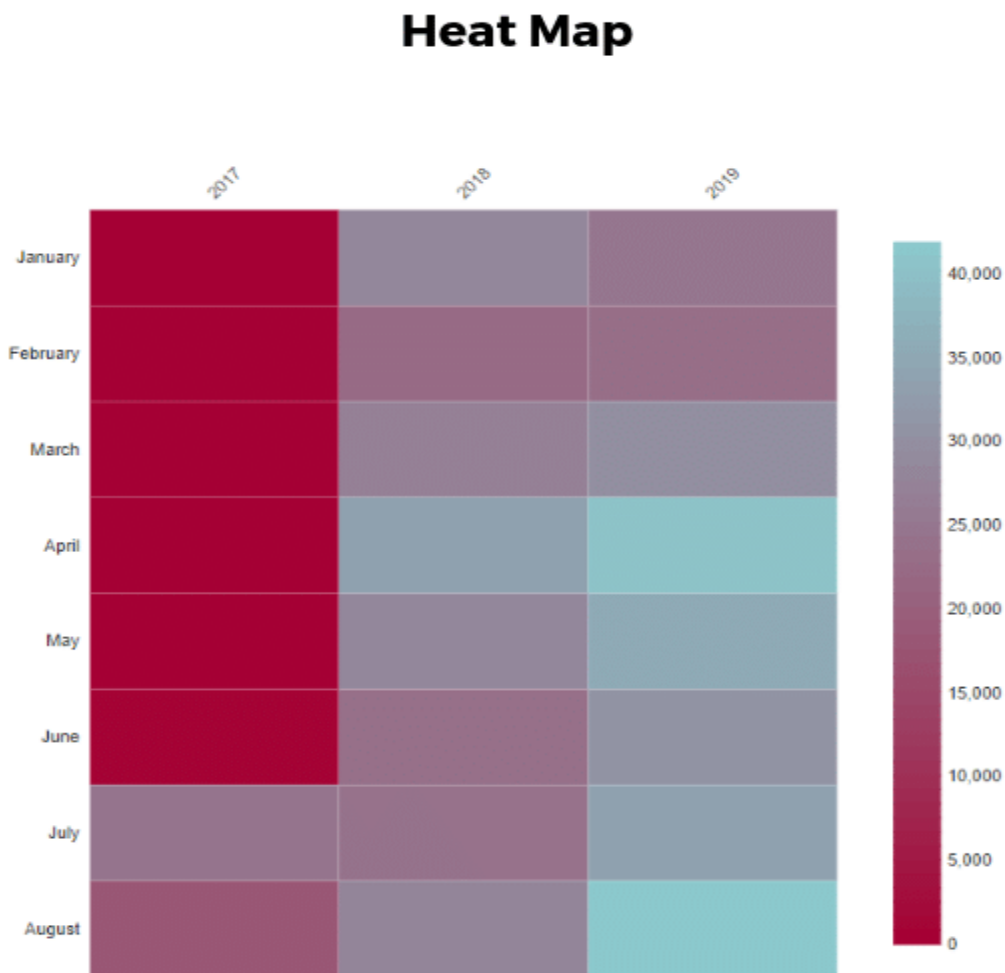
- **When to Use:** Scatter plots are often used in scientific research and data analysis to explore relationships between variables, such as height versus weight, marketing spend versus sales revenue, or age versus income.
- **Pros:** Good for correlation and clustering insights.
- **Cons:** Difficult with large datasets; overplotting.



## 1.6 Heatmap

A heatmap uses color gradients to represent data values across a two-dimensional space, making it easy to see patterns. Darker or lighter colors indicate higher or lower values (including positive and negative values), respectively, **making trends or concentrations easily visible**.

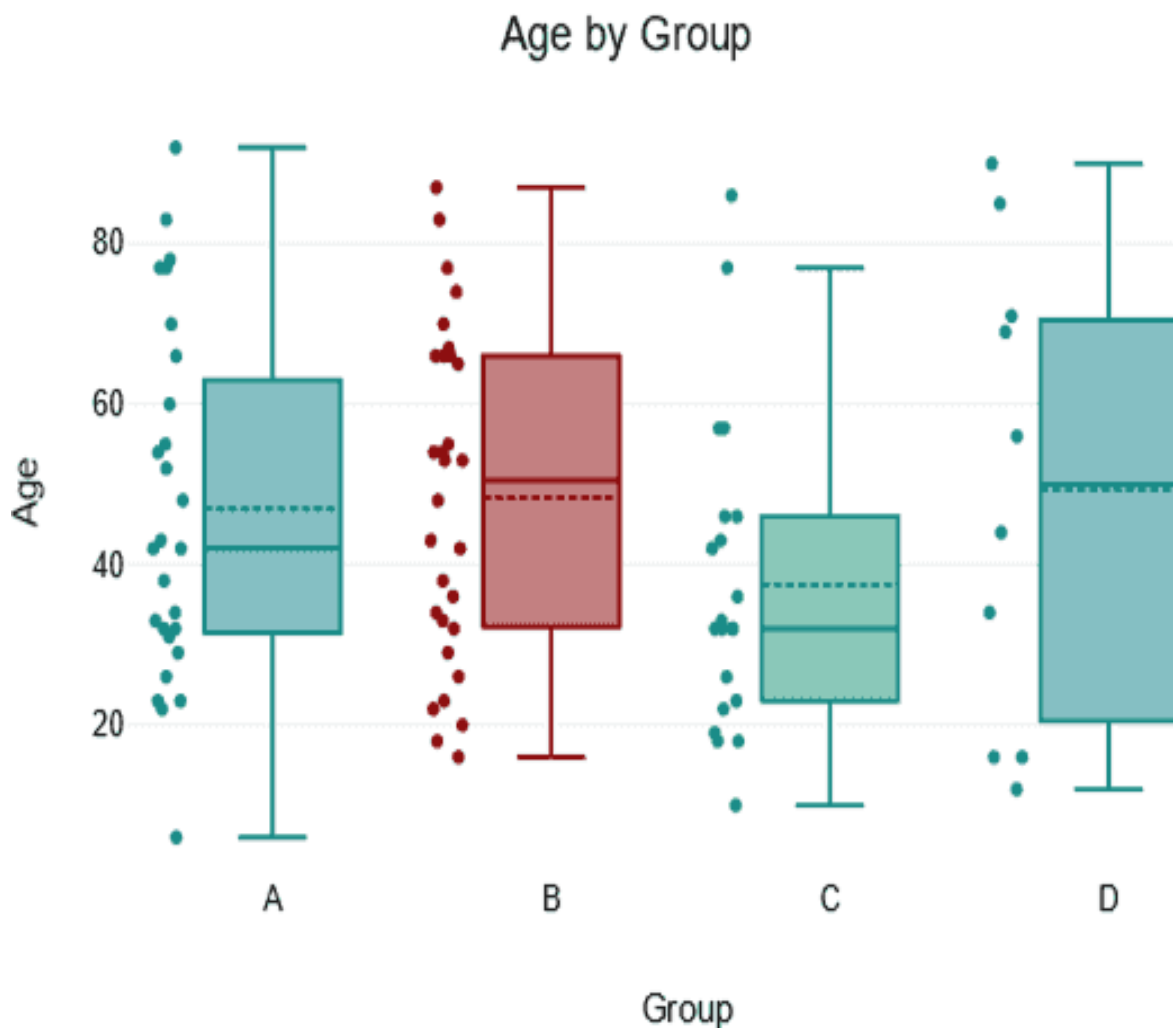
- **When to Use:** Heatmaps are great for showing data density or intensity, such as website user activity, sales performance across regions, or temperature variations on a map. It is also useful in correlation analysis and portfolio analysis.
- **Pros:** Quick insights from color intensity.
- **Cons:** Can be hard to interpret without clear scale.



## 1.7 Box Plot

A box plot uses boxes and whiskers to provide a visual summary of data distribution. The position of the box and whisker ends on the chart shows the areas where a significant portion of data lies. It is beneficial for **showing the spread of data and identifying any anomalies**.

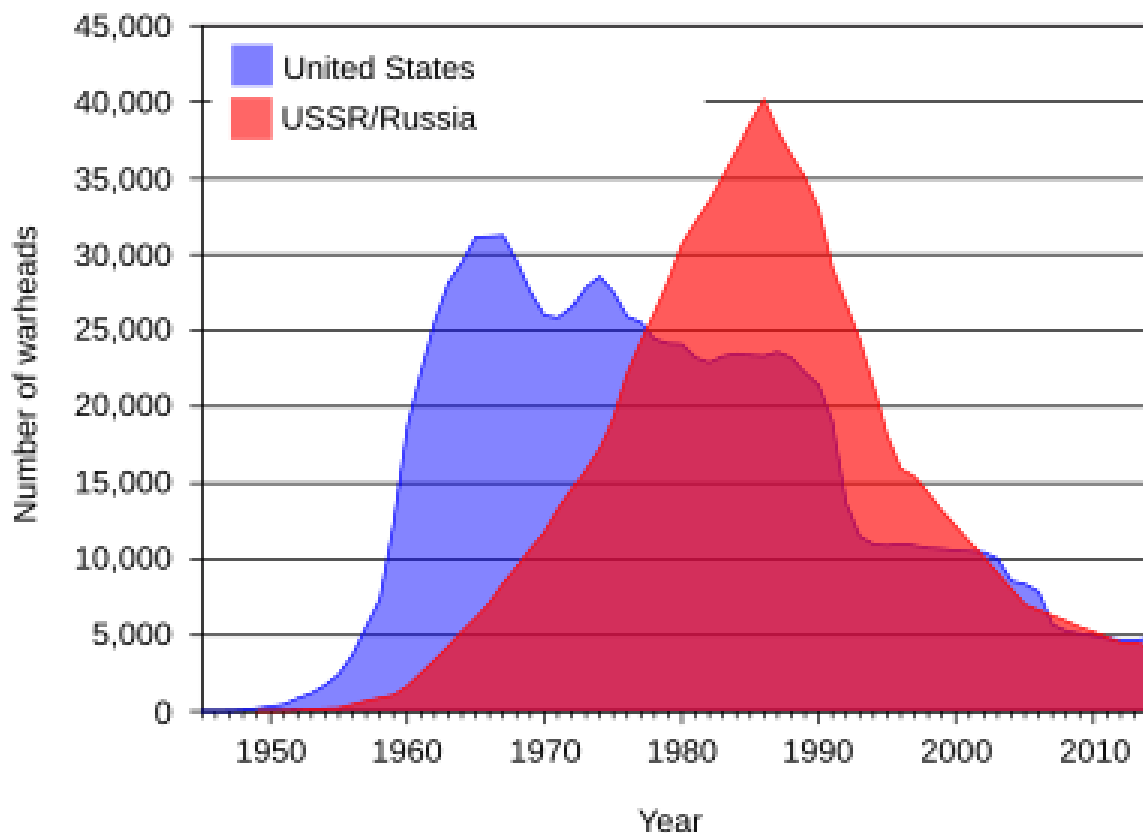
- **When to Use:** Box plots are commonly used in statistics to compare values of distributions between various groups. They are also useful for visualizing the range and variability in data like test scores, financial performance, or customer ratings.
- **Pros:** Highlights medians and outliers.
- **Cons:** Not intuitive for everyone.



## 1.8 Area Chart

An area chart is like a line chart, except the area under the line chart is filled in with color. This makes it easier to see the data's size and how it changes over time. It's **good for showing cumulative totals**. You can also use an area chart to **compare data from multiple categories**.

- **When to Use:** Area charts are often used to show how numbers grow over time, such as tracking total sales or revenue. They can also compare data points across multiple categories, such as sales across different products, to find the overall trend.
- **Pros:** Emphasizes magnitude.
- **Cons:** Can be hard to read if overused.

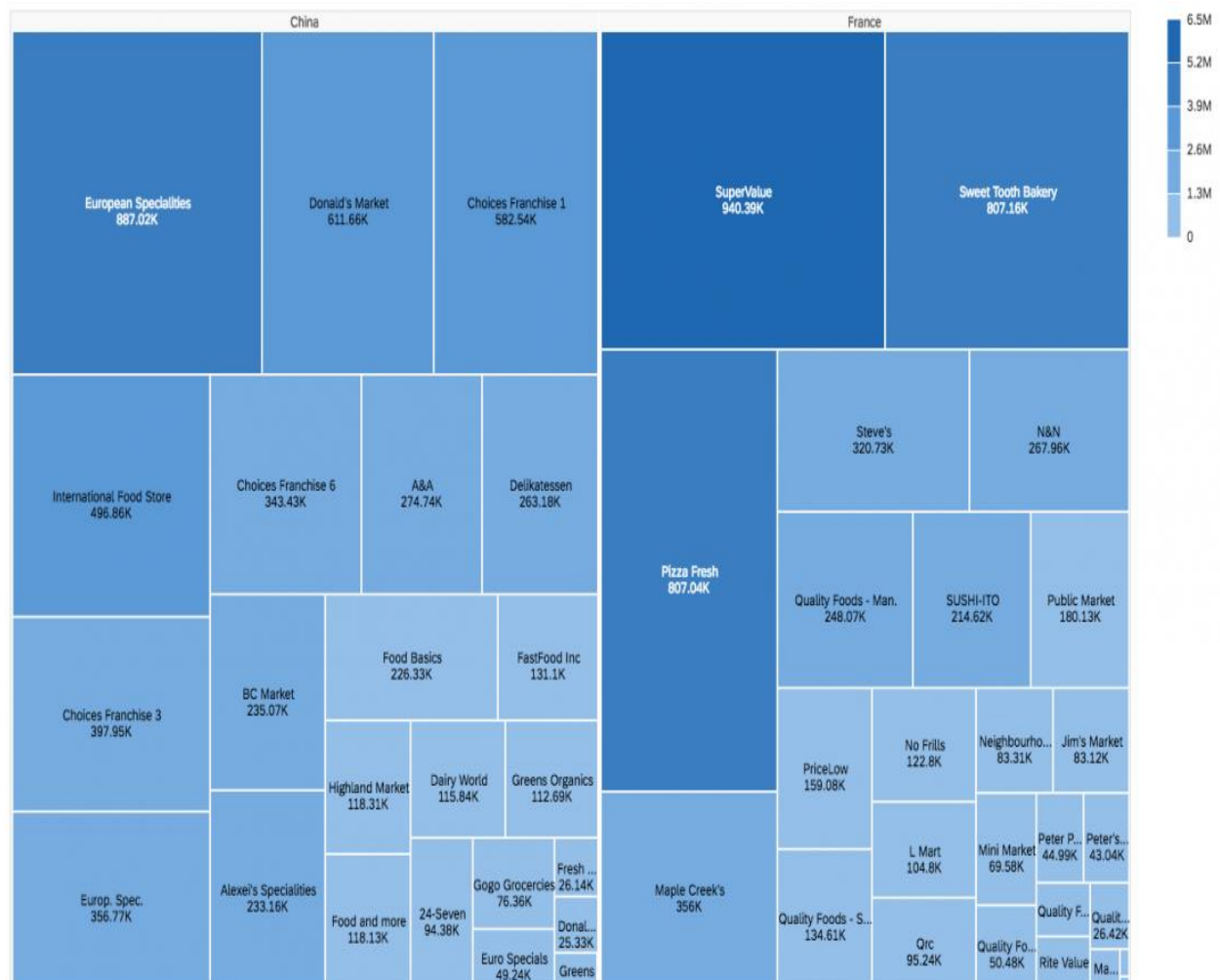




## 1.9 Treemap

A treemap chart **shows hierarchical data** using nested rectangles. Each rectangle represents a category, and its size reflects its value relative to the whole. This helps visualize the proportion of each item within a hierarchy.

- **When to Use:** A treemap chart **shows hierarchical data** using nested rectangles. Each rectangle represents a category, and its size reflects its value relative to the whole. This helps visualize the proportion of each item within a hierarchy.
- **Pros:** Compact and space-efficient.
- **Cons:** Hard to interpret exact values.



## 2. Languages for Data Visualization

- **Python:** Libraries like Matplotlib, Seaborn, Plotly, Bokeh.
  - **R:** ggplot2, plotly, lattice.
  - **JavaScript:** D3.js, Chart.js, Highcharts.
  - **SQL (via BI Tools):** Simple charts within tools like Power BI, Tableau.
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## 3. Tools for Data Visualization

- **Tableau:** User-friendly, interactive dashboards, drag and drop interface.
  - **Power BI:** Integrates well with Microsoft products, strong for enterprise use.
  - **Google Data Studio:** Free, web-based, integrates with Google services.
  - **Excel:** Great for quick and basic visualizations.
  - **Looker:** Cloud-based BI, integrates with modern data stack.
  - **Python/R:** Ideal for custom, reproducible, and complex visualizations.
  - **D3.js:** Highly customizable visualizations for web applications.
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## 4. Choosing the Right Visualization

- Use **bar/column charts** for categorical comparisons.
  - Use **line/area charts** for trends over time.
  - Use **scatter plots** for correlation.
  - Use **pie/treemap** for proportions.
  - Use **heatmaps/box plots** for distribution insights.
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## 5. Final Tips

- Always label your axes and legends.
  - Keep visualizations clean and uncluttered.
  - Use color wisely and consistently.
  - Know your audience and their needs.
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